Manuscript 14: Lean 4 and Arb certification record

Magnetic properties of the Eulerian profile at the special point $i/\sqrt{2}$, version 14.118

Roberto Alfano, 8 November 2026

1. Scope.

This record states, for each numbered result of manuscript 14, version 14.118, and for the computed values added to its text on 8 November 2026, what is kernel-checked in Lean 4, which scalar values are enclosed by Arb at 2000 bits, which steps are classical inputs and which formalisation tasks remain open. Equation, section and result numbers are those of version 14.118, identical to those of version 14.105: up to (141) they are those of version 14.91, and from (142) on they exceed those of version 14.91 by one, because version 14.95 added the line (142) with fifty decimals of the root x_0 of (139). All certificates are in the main branch of the repository `https://github.com/ralfanop/Certifications_Eulerian_polar-curve_computations` at commit `4909a891`, and were audited from 5 to 8 November 2026 (those of versions 14.95–14.105 in PR 63, those of version 14.118 in PR 64); each pull request (PR) below links its workflow runs. The Lean certificates use Lean `v4.35.0-rc3` with the pinned Mathlib dependency recorded in their workflows, compiled from source; every axiom audit lists only `propext`, `Classical.choice` and `Quot.sound`, with no `sorryAx` and no `Lean.ofReduceBool`. The numerical certificates use `python-flint 0.9.0`, exact rational arithmetic and Arb balls at 2000 bits; they certify the stated scalar quantities and do not prove infinite-dimensional operator identities.

2. Overview.

In Table 1, "yes" means that the stated content is certified, apart from the classical inputs listed in Section 4, and "—" that it is certified numerically only.

Table 1. Certification status of the numbered results.

Result	Lean 4	Arb	Pull requests
Proposition 2.1 (special-value chain)	—	yes	PR 38
Section 2.1 (discriminant of F_n)	yes	yes	PR 62
Corollary 2.2 (rational bounds)	—	yes	PR 38
Lemma 2.3 (uniqueness of z_0)	yes	yes	PR 44
Proposition 3.1 (Sonine)	yes	yes	PR 31, PR 38
Proposition 3.3 (modified Helmholtz)	yes	yes	PR 38, branch `cor36`
Proposition 4.1 (Langevin)	yes	yes	PR 38, PR 46
Theorem 4.2 (noninteracting magnet)	yes	yes	PR 40, PR 48, PR 61
Section 4.5 and its Table 1 (superparamagnetic fields)	—	yes	PR 64
Figure 1 (caption) and the paragraph of Figure 2	yes	yes	PR 64
Figures 1–3 (plotted data)	—	yes	PR 64
Displays (54)–(55) (Riccati, Painlevé III)	—	yes	PR 61
Display (60) (n-vector)	$\lambda = 0$	$\lambda = 0, \pm\frac{1}{2}$	PR 40, PR 47
Definition 5.1 (Gram operator)	yes	—	PR 33
Theorem 5.2 (Gram commutation)	yes	—	PR 37

Result	Lean 4	Arb	Pull requests
Corollary 5.3 (Eulerian specialisation)	yes	yes	PR 33, PR 38
Proposition 5.4 (spheroidal basis)	yes	—	PR 41, PR 42, PR 43
Proposition 6.2 (two determinants)	yes	yes	PR 44, branch `dirichlet-finite`
Proposition 6.3 (resolvent traces)	yes	yes	PR 44
Corollary 6.4 (Hilbert–Schmidt trace)	yes	yes	PR 44
Proposition 6.5 (zeta expansion)	—	yes	PR 39
Theorem 7.1 (modular duplication)	yes	yes	PR 44
Theorem 8.1 (quadratic covering)	yes	yes	PR 45
Proposition 8.2 (Montgomery–Taylor)	yes	yes	PR 45, PR 61
Proposition 8.4 (Dirichlet L-series, self-dual coupling)	yes	yes	PR 45, PR 63
Proposition 8.6 (paths and Viète)	yes	yes	PR 45, PR 61, PR 62, PR 63
Proposition 8.7 (classical PT-symmetric oscillators)	yes	yes	PR 61, PR 63
Text after Proposition 8.7 (discriminants)	yes	yes	PR 62, PR 63
Proposition 8.8 (lattice walks)	yes	yes	PR 45
Lemma 9.2 (time reversal)	yes	yes	PR 45, PR 62
Theorem 9.3 (Bloch antipode)	yes	yes	PR 45, PR 46, PR 62
Corollary 9.4 (averages over pure states)	yes	yes	PR 45, PR 46
Proposition 9.5 (Brillouin splitting)	yes	yes	PR 44, PR 45
Text after Proposition 9.5 (electron spin)	—	yes	PR 62
Proposition 9.7 (involutions)	yes	yes	PR 45
Theorem 10.1 (Eulerian–Parry)	yes	yes	PR 30

3. Content of the certificates.

Sections 2–3. The script `special_point_i_sqrt2_arb.py` encloses $r_E(z_0) = \sinh(a)/a$ and checks, in exact rational arithmetic, the bounds (19). The duplication identity (15) is kernel-checked in the `Bridge` library. The Sonine certificates (PR 31; CI 41853697588) prove the integral moments c_n for every index, their factorial formula, the interchange of the integral with absolutely convergent even series, and the spherical integral identities of the series-defined J_0 and I_0 at every complex argument; the radial-series certificate (CI 41852735785) proves the intertwining for coefficient-defined radial operators under its summability hypothesis. The differential equation (36) is kernel-checked in six declarations on the branch `manuscript14-cor36-lean-arb-20261104` (CI 41824974185).

Section 4. The values (52)–(53) are enclosed and proved correctly rounded (CI 42207399251). The module `BlochMeasureCertificate` (PR 46) proves $\int e^{x\omega_3}\,\mathrm{d}\mu = \sinh x/x$ with ω_3 uniform on $(-1,1)$, which gives the normalisation (38) through the coordinates of the first input of Section 2.2. The module `Theorem42Certificate` (PR 48; CI 42356565741; twenty-one audited statements) proves Theorem 4.2: for (i), $u\,\partial_u \log B(u) = [1 + L(H/2)]/2$, its derivative $L'(H/2)/4$ in H, the mean of the quasi-power $B(u)^{n+1}$, and the mean and variance of any finite law tilted by u^k as the first two derivatives of its logarithmic normalisation; for (ii), the probabilities $1/2$, $1/6$ and $1/4$ of a descent, of two adjacent descents and of two descents at distance at least two, for every n, hence (46), the mean $(n-1)/2$, (47) and (50); for (iii), the factorisation into independent spins with the fields (51); and (iv). The script of PR 40 (CI 42239016609; 145 checks) certifies (46)–(47) and all

descent correlations for $n \le 9$, (47) for $n \le 670$, (50) with its coefficients, (57)–(59) at a and b, the real zeros and (51) for $n \le 18$, and (60) at $\lambda = \pm 1/2$. The module `PlanarSpinCertificate` (PR 47; CI 42355540276; eight audited statements) proves (60) at $\lambda = 0$: the arcsine law ν_0 is the image of the uniform angle under cos, its moments are $\binom{2m}{m}/4^m$ and 0, and its transform is $\sum_k (x^2/4)^k/(k!)^2$, also as an average over the full circle; its script (CI 42355345132; 13 checks) compares the series with I_0 at ten points, among them $I_0(z_0) = J_0(1/\sqrt{2})$, and with the integrals.

Section 5. CI 41840633678 checks seven declarations on the complex-L^2 Gram integral, its exact parameter and measure, the nuclear decomposition and the basis-independent trace, and CI 41841459971 thirteen kernel and graph-domain lemmas; the kernel balance, the core-extension criterion, the Gegenbauer spectrum and the trace formula (64) are in the `GegenbauerFock` library. The module `GramCommutationTheorem52` (PR 37; CI 42209958304) proves Theorem 5.2 for every $c > 0$, the inclusion $\mathcal{S}_{1/2,c}\mathcal{H}_{1/2} \subset \mathrm{Dom}(\mathcal{A}_c)$ and (63), for the graph closure of the polynomial Legendre operator perturbed by M_{ξ^2}. The module `GegenbauerSpectralRealisation` (PR 42; CI 42260542889) identifies this closure with (8)–(9), proves it selfadjoint with compact resolvent, and proves that $\mathcal{A}_c + c^2 + 1$ has a compact inverse. The module `SpheroidalProposition54` (PR 41; CI 42249877445) proves Proposition 5.4: parity, simplicity within each parity class (from the moments rather than from (74)), positivity of the $\mathcal{S}_{1/2,c}$-eigenvalues and a joint Hilbert basis with $\sum_n \mathrm{g}_n = {}_1F_1(1/2; 3/2; c)$. The module `SpheroidalDLMFIdentification` (PR 43; CI 42272154201) identifies the basis vectors with the spheroidal functions $\mathrm{Ps}_n^0(\cdot, -c^2)$, the definition of DLMF §§30.2–30.3 entering as an explicit hypothesis.

Section 6. The branch `manuscript14-dirichlet-finite-20261104` (CI 41824926567, CI 41825497267) checks fourteen declarations on the finite truncations of the products and traces, with thirteen Arb comparisons. The module `Section67ScalarCertificate` (PR 44; CI 42283397911; nineteen audited statements) proves the Euler products (13)–(14) as the products (79)–(80) over the spectra (76) and (78), with (81), the spectral sums (82)–(84) and (89)–(90) for every $x > 0$, the values (85) and (87), and (91)–(92). The script of PR 39 (CI 42224556937; 92 checks) certifies Proposition 6.5: the Bernoulli numbers and the rational terms of (97), the values $\zeta(2m)$ with exact tail bounds, the termwise agreement of the two forms of (94), and the enclosure of $\log r_E(z_0)$.

Section 7 and Lemma 2.3. The same module proves Lemma 2.3 (uniqueness of z_0 in the upper half-plane, $S(z_0) = z_1 = 2z_0$, $S(z_1) = z_0$ and (22)–(23)) and Theorem 7.1: (98), the product (99) over the spectrum of A_2 and its splitting into the products over the spectra of A and A_{DN}, (100) in product form, (101)–(103) and the companion values. Its script (CI 42282862670; 45 checks) certifies (21)–(103) as ball identities, the products and traces with rigorous tails, and the rounding of the values of Remark 7.4.

Sections 8–9. The module `Section89Certificate` (PR 45; CI 42299447127; thirty-three audited statements) proves: in Theorem 8.1, the fibres $\{\tau, -\tau\}$ of $\mathcal{Q}$, the deck transformations, (115) with r_E the limit of the partial products of (114), $m(-H) = 1 - m(H)$, the lifts of (116), the doubling of winding numbers, the fibres over $\pm 1/4$, (118) and the uniqueness of z_0; Proposition 8.2 ((121) from the Mittag-Leffler expansion of the cotangent, the logarithmic derivative, (123)); in Proposition 8.4, (126), (131)–(135), $4G_*^2 + 4G_* - 1 = 0 \Leftrightarrow \sinh 2K_{\mathrm{sd}} = 1$ and the arcsine moments; part (i) of Proposition 8.6, (136) with its zeros, the Perron eigenvalue, the eigenvector relation and (137); the two identities in the proof of Proposition 8.8; Lemma 9.2; $\omega_{\mathrm{Bl}} \circ \Theta = -\omega_{\mathrm{Bl}}$ and $|\omega_{\mathrm{Bl}}| = 1$; (161), (162) and the time reversal of the thermal state, the spectral forms (159)–(160) being (80) and (83) of PR 44; and parts (i), (iii) and (iv) of Proposition 9.7. The module `BlochMeasureCertificate` (PR 46; CI 42307987424; fifteen audited statements) proves that the image of μ under $\mathfrak{z} \mapsto (\omega_3, \arg \mathfrak{z})$, the axial coordinate and the azimuth of $\omega_{\mathrm{Bl}}(\mathfrak{z})$, is $(4\pi)^{-1} \mathrm{d}\omega_3 \, \mathrm{d}\varphi$ on $(-1, 1) \times (-\pi, \pi)$, that μ is

a probability measure, (156) for every real $x \neq 0$ and (157), the mean $L(x)$ of ω_3 under $e^{x\omega_3}\mathrm{d}\mu$, the invariance of the law of ω_3 under $\mathfrak{z} \mapsto -1/\bar{\mathfrak{z}}$, and the bijection of $\mathbb{C}$ onto the sphere without its south pole. The script of PR 45 (CI 42299795864; 115 checks) certifies the rounding of (120), (125) and (127), the closed forms as ball identities, the traces of (121) and (123) with two-sided tail bounds, (122) in exact arithmetic, (128)–(130), the characteristic polynomials of the paths for $m \leq 12$, (138), (149) for $2n \leq 24$, (150)–(151) at $q = 1/8$, (156)–(157) as integrals, and (158) against the sublevel sums for $J = 1/2, 1, 3/2, 7/2$.

Additions of 7 November 2026 (evening). The script `manuscript14_evening_additions_arb.py` in `numerics/arb` (PR 61; CI 42422614198; 45 checks, exact rational and $\mathbb{Q}(\sqrt{2})$ arithmetic and Arb at 2000 bits) certifies: the Riccati equation (54) of Section 4.3 at six points, its general solution L_ϱ for four values of ϱ, the saddle (eigenvalues 1, -2, separatrix slope $1/3 = L'(0)$) and the node (eigenvalues 1, 2) of the Riccati foliation, $\mathsf{w}(z) = 1/z - \cot z$ with $\mathsf{w}(ix) = iL(x)$ and the third Painlevé equation (55) at four complex points; in the proof of Theorem 4.2, Knuth's Bernoulli numbers ($B_1 = 1/2$), the equality of the kernel (49) with the form of Borodin, Diaconis and Fulman, the generating function (48), the mean $(n-1)/2$ and the variance $(n+1)/12$ for $n \leq 30$ and, with $\mathrm{k}(0) = -1/2$, the mean $-(n-1)/2$, the negative variance and the polynomials $3 - u$ and $u^2 - 8u + 13$, the Eulerian polynomials as $n!\det[I + (x-1)\mathcal{K}_{n-1}]$ for $n \leq 7$, and Bernoulli's sums of powers with $B_1 = \pm 1/2$; the eight terms of (122) and their zeta form; the middle term of (138); and Proposition 8.7: the roots of $32x^4 - 16x^2 + 1$, which is 32 times (139), its irreducibility (Eisenstein at 2 for $\mu_{\mathrm{hex}}^4 - 4\mu_{\mathrm{hex}}^2 + 2$ and the criterion for biquadratics) and primitivity, so that it is the minimal polynomial of x_0 over $\mathbb{Z}$, (140), the integral (141), Bender's polynomial (143)–(146) for seven rational values of $\gamma_{PT}^2 \in [0, 2)$, (147) with $\cot(\pi/8) = 1 + \sqrt{2}$, (148), and the midradius of the square antiprism with unit edges. The module `QuarticResidueCertificate` (PR 61; CI 42422614173; fourteen audited statements) proves the algebraic identities of (139), (140), (143)–(148) for $x_0 = \sqrt{2 + \sqrt{2}}/\sqrt{8}$ and $x_1 = 1/(2\sqrt{2 + \sqrt{2}})$, with (139) in the integer form $32x^4 - 16x^2 + 1$; the integral (141), the irreducibility and the value of $\cot(\pi/8)$ are certified only in the script.

Additions of 8 November 2026 (version 14.92). The script `manuscript14_v1492_additions_arb.py` in `numerics/arb` (PR 62; CI 42478168014; 30 checks, exact integer, rational and Gaussian-rational arithmetic and Arb at 2000 bits) certifies: in Section 2.1, that the discriminant of $F_n(t) = t^n - t^{n-1} - 1$, in the normalisation $R(f, f')/a_n$ of Gelfand, Kapranov and Zelevinsky (Chapter 12, Eq. (1.29)), is $(-1)^{n-1}[n^n + (n-1)^{n-1}] \neq 0$, by Sylvester determinants for $n = 2, \ldots, 40$ and by their trinomial formula (1.38) for $n = 2, \ldots, 300$, the layout being checked against their formulas (1.33)–(1.35) on random polynomials; after Proposition 8.7, the discriminant 2^{23} of $32x^4 - 16x^2 + 1$, which is 32 times (139), the discriminant $16(1-g^2)[4(1-g^2) - (2-\gamma^2)^2]^2$ of Bender's polynomial (143) for 49 rational pairs, its value $s^{12}/128 > 0$ on (145)–(146) and its zeros where two frequencies coalesce or one vanishes; after Proposition 8.6, the dispersion relation of the lossless line of loops and $2\cos(\pi/8) = \mu_{\mathrm{hex}}$; after Proposition 9.5, the rounding of $a\,k_{\mathrm{B}}/\mu_{\mathrm{B}} \approx 0.526$ T/K, of 3.80 K at 2 T, of $e^{1/\sqrt{2}} \approx 2.028$ and of 3.88 K, with $\mu_{\mathrm{B}} = 9.274 \times 10^{-24}$ J/T and with the CODATA value; in Lemma 9.2, that $-i\sigma_2$ is the quaternion unit e_2 of the representation of Pandey, Kumar and Puri; in Theorem 9.3, that the Hopf projection $(2\overline{z_1}z_2, |z_1|^2 - |z_2|^2)$ is the Bloch map (152). The module `V1492AdditionsCertificate` (PR 62; CI 42478417105; nine audited statements) proves the trinomial formula at the parameters of F_n and its nonvanishing, the even case of the quartic formula, the discriminants of $32x^4 - 16x^2 + 1$ and of (143), the value on (145)–(146) and its positivity, and the factorisations at coalescing and vanishing frequencies; the formulas (1.35) and (1.38) themselves enter as classical inputs.

Changes of versions 14.95–14.105 (8 November 2026). The script `manuscript14_v14104_`

`additions_arb.py` in `numerics/arb` (PR 63; CI 42487059795; 34 checks, exact rational and $\mathbb{Q}(\sqrt{2})$ arithmetic and Arb at 2000 bits) certifies: the fifty decimals of (142), exact by truncation (the fifty-first is 4), with x_0 computed by four routes, $\mu_{\mathrm{hex}}/\sqrt{8}$, the integral of (141) by rigorous Arb integration, $\sqrt{1+1/\sqrt{2}}/2$ and $1/(4\sin(\pi/8))$; the monic form (139), whose 32-fold is $32x^4-16x^2+1$, its roots $\pm x_0$, $\pm x_1$, (140), (147) and its discriminant $1/128 = 2^{23}/32^6$, by the Sylvester determinant, by the quartic formula (1.35) and by the squared differences of the roots; the forms $2(2-\gamma_{PT}^2)$ of (145) and $1-[(2-\gamma_{PT}^2)/\sqrt{8}]^2$ of (146), (144) as $\mathsf{P}_{PT}(s_{PT}x) = s_{PT}^4$ times (139) for seven rational values of $\gamma_{PT}^2 \in [0,2)$, the discriminant $s_{PT}^{12}/128$ of P_{PT} as s_{PT}^{12} times that of (139), the frequencies $s_{PT}x_k$ and (148); (126)–(127); the fraction form of (137), the first three factors $\cos(\pi/4)\cos(\pi/8)\cos(\pi/16)$ of Viète's product; and, for (128)–(129), $\zeta(1/2) < 0$, the value $L_2(1/2,\chi_1)$ as $2^{-1/2}\zeta(1/2,1/2)$ through the Hurwitz zeta function and as Arb's Dirichlet L-function of the principal character modulo 2, both equal to $(1-2^{-1/2})\zeta(1/2)$ as in Table 22 of Mathar (whose entries $s = 2,3,4$ are reproduced), hence $1 - L_2(1/2,\chi_1)/\zeta(1/2) = 1/\sqrt{2}$ and $1/2 - L_2(1/2,\chi_1)/\zeta(1/2) = (\sqrt{2}-1)/2$, the integral of (128), and the first thirty digits of $1+G_* = (1+\sqrt{2})/2$ against OEIS A174968. The module `V14104AdditionsCertificate` (PR 63; CI 42487059874; eleven audited statements) proves that 32 times (139) is $32x^4-16x^2+1$, the discriminant $1/128$ of (139) and its relation to 2^{23}, that $x_0 = \sqrt{1+1/\sqrt{2}}/2$ is a root of (139), the forms (145)–(146), (144) under (145)–(146), the discriminant of P_{PT} on (145)–(146) as s_{PT}^{12} times that of (139), (148), the fraction form of (137), and (128)–(129) from the identity $L_2(s,\chi_1) = (1-2^{-s})\zeta(s)$ and $\zeta(1/2) \neq 0$, taken as hypotheses.

Changes of version 14.118 (8 November 2026). The script `manuscript14_v14118_section45_figures_arb.py` in `numerics/arb` (PR 64; CI 42505194706; 132 checks, exact rational arithmetic and Arb at 2000 bits) certifies: in Section 4.5, $\mu_{\mathrm{p}} = 2.85\,\mu_{\mathrm{B}}$ and $x \approx 0.0065$ at 1 T for the oxygen molecule at 293 K, the inductions of about 54 T and 108 T at H_0 and H_1 and of about 246 T at the edge $|H| = 5/2 + 1/\sqrt{2}$ of the window of Figure 1(b), those of about 5.3 mT and 10.5 mT for the cobalt particle at 300 K and below 0.075 T up to $|H| = 10$, and $\ln(10^{11}) \approx 25$; the twenty entries of Table 1; in the paragraph of Figure 2, the centres $m(H_0) = 0.44156$, $m(H_1) = 0.38590$ and the widths $L'(a)/4 = 0.08129$, $L'(b)/4 = 0.07562$ of the histograms for every $n = 10,\dots,670$, the only inflection point of (44) at $H = 0$ ($L'' < 0$ on $(0,\infty)$, by an exact Bernoulli bound on $(0,1/2]$, Arb bisection on $[1/2,5]$ and an exact exponential bound on $[5,\infty)$, with L'' odd) and the convexity at H_0, with second derivative $-L''(a)/8 > 0$; in the caption of Figure 1, $m(H) = 1 + 1/(e^H-1) - 1/H$, $m(-H) = 1 - m(H)$, the approach to the saturation values as $1/|H|$, the window centred at H_0, and the frame, guides, markers, labels and tangent of the figure source; every plotted point of Figures 1–3: the 2001 and 1601 points of the two panels of Figure 1, the 4008 centres and widths of Figure 2 for $n = 3,\dots,670$, correctly rounded from the exact Eulerian numbers, with the top vertex $p_{3,2}(1) = 2/3$, and the 536 remainders of Figure 3 within 10^{-12} with every floor decided, the Parry roots being isolated by exact sign changes, with $h_E \approx 0.9961$ and the endpoint and the left limit at β_{31}; and the 52 decimals of $\mathrm{Tr}(2/(2A+I))$ in (112), Remark 7.4. The module `V14118Section45Certificate` (PR 64; CI 42505444820; nine audited statements) proves that L is odd, $m(-H) = 1 - m(H)$, $\coth(H/2) = (e^H+1)/(e^H-1)$, $m(H) = 1 + 1/(e^H-1) - 1/H$, the bounds $1 - 1/H < m(H) < 1$ for $H > 0$ and $0 < m(H) < -1/H$ for $H < 0$, $H[1-m(H)] = 1 - H/(e^H-1)$, and the limits $H[1-m(H)] \to 1$ as $H \to +\infty$ and $-H\,m(H) \to 1$ as $H \to -\infty$. Version 14.118 corrects two sentences of Section 4.5 (the window of Figure 1(b) at about 246 T; the whole curve of Figure 1(a), up to $|H| = 10$, below 0.075 T) and the sign of the second derivative at H_0 in the paragraph of Figure 2.

Section 10. Theorem 10.1 is kernel-checked in full (CI 41812736721): the adjacent thresholds, the greedy digits, the rational orbit-defect bounds, the endpoint jump, the rational series of

$\beta_E = \sqrt{2}\sinh(1/\sqrt{2})$ (module `EulerianSinhSeries`), its transcendence and the absence of eventual periodicity; the finite word at β_{31} is checked separately. The Hermite–Lindemann instance is compiled from source adapted, under its Apache 2.0 licence and with attribution, from Yuyang Zhao's Mathlib pull request #28013, not yet merged on 7 November 2026.

4. Classical inputs.

The following enter the certificates as explicit hypotheses or are not reproved: the inputs of Section 2.2 of the manuscript, among them the surface measure $\mathrm{d}\xi\,\mathrm{d}\varphi/(4\pi)$ of $\mathbb{S}^2$ in axial coordinates; the identification of the Fredholm determinant and trace of a positive trace-class operator with the product and sum over its spectrum; the quasi-power approximation $\mathrm{A}_n(u) = B(u)^{n+1}[1+O(|H/(H+2\pi i)|^n)]$ with $u = e^H$ (Flajolet–Sedgewick, Example IX.37); the real, simple, negative zeros of A_n (Frobenius); the determinantal structure of the descent process (Borodin–Diaconis–Fulman, Theorem 5.1); the j-values of Lemma 2.3; the covering-space statements of Theorem 8.1(i) and the exact sequence (117); the definition of the spheroidal functions in DLMF §§30.2–30.3; the discriminant formulas (1.29), (1.35) and (1.38) of Gelfand, Kapranov and Zelevinsky, Chapter 12; and the identity $L_2(s,\chi_1) = (1-2^{-s})\zeta(s)$ of Table 22 of Mathar, with $\zeta(1/2) \neq 0$, both checked in Arb at $s = 1/2$. The results listed as open formalisation tasks in the earlier versions of this record are classical and enter as inputs as well: the behaviour at the origin of the radial operators H_2, H_3 on even entire functions and the uniqueness of the regular solution of the radial equation, from the theory of regular singular points (Propositions 3.1 and 3.3); Archimedes' theorem, the uniform law on $[-1,1]$ of the axial component of a uniform point of $\mathbb{S}^2$ (Proposition 4.1); the identification of ratios of zeta-regularised determinants of Sturm–Liouville operators with Fredholm determinants, of Gel'fand and Yaglom in the form of Kirsten and McKane (Proposition 6.2), and the expansion of $\log(\sinh x/x)$ in the even zeta values from Euler's product (Proposition 6.5, whose values are certified in Arb); the decomposition of $L^2(0,2)$ into the parts odd and even under the reflection $t \mapsto 2-t$ (Theorem 7.1); Abel's theorem on power series at the boundary (the value (128) of Proposition 8.4); the spectrum $2\cos(j\pi/(m+1))$ of the path of m sites, with characteristic polynomial $U_m(y/2)$, and Euler's beta integral $\int_0^\infty y^{v-1}(1+y)^{-1}\,\mathrm{d}y = \pi/\sin(\pi v)$, $0 < v < 1$ (part (ii) of Proposition 8.6, (138), and the integral (141) of Proposition 8.7); Eisenstein's criterion (the irreducibility in Proposition 8.7); the Neumann series of the resolvent of the adjacency operator of $\mathbb{Z}^2$ for $|q| < 1/4$ and Pólya's generating function of returns (Proposition 8.8); the stereographic homeomorphism of $\mathbb{C}\cup\{\infty\}$ onto $\mathbb{S}^2$ (Theorem 9.3 and part (ii) of Proposition 9.7); the invariance of the Fubini–Study measure under the antipodal map (Corollary 9.4); and Lazarević's inequality $(\sinh x/x)^3 > \cosh x$ for $x \neq 0$, which is the sign of L'' on $(0,\infty)$ and hence the single inflection point of Figure 1, certified also in Arb. The statements of the additions of version 14.92 that are quotations of the literature (the line of loops of Solymar and Shamonina, Dyson's classes, the Berry phase and the Hopf bundle) are not reproved.

5. Open formalisation tasks.

None. The statements listed here in the earlier versions of this record are classical results and enter as the inputs of Section 4. For Proposition 2.1 and Corollary 2.2 this record lists only the numerical certificates, and the data of Figures 1–3 are certified as numbers, in Arb.

6. Numerical jobs.

The workflow `manuscript14-arb-2000bit.yml`, introduced by PR 38, runs the fourteen numerical scripts of the manuscript at 2000 bits: 572 checks in all, of which 162 in the nine original scripts (63 of the register of 4 November 2026, 35 Bessel, 15 Gram and resolvent, 13 Dirichlet, 28 Sonine and modified Helmholtz, 8 for Proposition 4.1), 92 for Proposition 6.5, 145 for Section 4.6, 45 for Sections 6–7, 115 for Sections 8–9 and 13 for (60) at $\lambda = 0$. The exact rational truncation bounds are those of the original scripts; only the ball precision was raised. The same workflow runs, since PR 61, PR 62 and PR 63, the certificates of the additions of 7 November 2026 and of versions 14.92 and 14.95–14.105 of 8 November 2026, with 45, 30 and 34 checks, and, since PR 64, the certificate of version 14.118, with 132 checks.